Automated Experiment Report

Run folder: run_20260419_061735

Prepared on 2026-04-19 02:26:51. This report was created automatically from saved run artifacts using deterministic Python rules only.

Model family	Horizons
Direct Hurdle Tabular Compact Feature Recipe	24, 48
Platform	**XGBoost device**
Linux-6.6.113+-x86_64-with-glibc2.35	cpu
Classifier configs	**Regressor configs**
2	2
Pairs per horizon	**Total fold fits**
4	8

Executive summary

- The run archived 143 tracked artifacts.
- Best 24h pair: h24__clf_logen01__reg_xgbr01.
- Best 48h pair: h48__clf_xgbc01__reg_xgbr01.
- For horizon 24h, the winning pair was Logistic Enet + Xgb Regressor. Average county RMSE expected was 82.0886; Improved by 4.6211 (5.33%). Stage-1 ROC AUC was 0.6919, indicating limited discrimination.
- For horizon 48h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 64.3431; Improved by 2.3737 (3.56%). Stage-1 ROC AUC was 0.8740, indicating strong discrimination.
- Submission-style CSVs were exported from: Best validation candidate.
- Demo-only evaluation files are present. They are useful for sanity checks only and are not the hidden challenge evaluation.

Configuration and modeling decisions

- Architecture: the run uses a two-stage direct hurdle pipeline ('direct_hurdle_tabular_compact_feature_recipe'). Stage 1 predicts whether outage is non-zero; Stage 2 predicts outage size conditional on positive outcomes only.
- Combination rule: the final expected outage equals stage-1 non-zero probability times the stage-2 positive prediction.
- Target transform: Stage 2 uses log1p on positive outages before fitting the regressor.
- Forecast horizons: 24h, 48h.
- Forecast-origin sampling: every eligible hour was used.
- Minimum training history before the first fold: 21 days (504 h).
- Maximum training history: capped at 28 days (672 h).

- Temporal validation design: manual aligned splits were specified - 24h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-12 00:00:00; 48h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-12 00:00:00.
- Target lag features: 1, 24 hours.
- Weather screening: up to 10 weather variables were allowed; 10 were retained in this run.
- Weather outlier handling: enabled with 'clip_to_nan_then_linear_interpolate', so extreme weather values were clipped rather than turned into missing values.
- Execution hardware: GPU acceleration was not available; the run executed on CPU.
- Prediction export policy: submission CSVs were generated from Best validation candidate.

Search space summary

component	families	config_count
Classifier	logistic_enet, xgb_classifier	2
Regressor	elasticnet_regression, xgb_regressor	2

Run inventory

The table below shows the top-level artifact folders detected in the run. The checkpoints folder is runtime infrastructure and may be intentionally absent from shared GitHub uploads.

subfolder	present	file_count	note
tables	Yes	34	
predictions	Yes	12	
models	Yes	40	
metrics	Yes	28	
plots	Yes	17	
checkpoints	Yes	10	Runtime-only resume artifacts; may be omitted from shared GitHub uploads.

Temporal evaluation design

The backtest table below maps the training and validation windows used for each horizon and fold.

horizon	fold_id	val_origin_idx	val_origin_timestamp	train_origin_start_idx	train_origin_start_timestamp	train_origin_max_idx	train_origin_max_timestamp	validation_window_start	validation_window_end
24	1	1,728	2023-06-12	1,464	2023-06-01	1,704	2023-06-11	2023-06-12 01:00:00	2023-06-13
48	1	1,728	2023-06-12	1,464	2023-06-01	1,680	2023-06-10	2023-06-12 01:00:00	2023-06-14

Feature frame and data overview

metric	value
rows_in_model_df	179,363
base_numeric_features	17
categorical_features	0.0000
feature_warmup_hours	24
max_horizon	48
selected_weather_feature_count	10

metric	value
mean_outage	45.2483
median_outage	0.0000
std_outage	452.2684
max_outage	23,346
zero_rate	0.7047
nonzero_rate	0.2953

Selected weather features retained in this run: 10.

feature	abs_corr_with_outage	signed_corr_with_outage
vstm	0.0539	-0.0539
unknown_2	0.0523	0.0523
veg	0.0522	0.0522
vvcsh	0.0488	-0.0488
unknown	0.0439	0.0439
v	0.0422	-0.0422
ustm	0.0415	-0.0415
mdens	0.0408	0.0408
vucsh	0.0365	-0.0365
lai	0.0350	0.0350

Candidate ranking by horizon

Rows are ranked within each horizon using average county RMSE expected first and overall RMSE expected second.

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
1	24	logistic_enet	xgb_regressor	82.0886	258.3323	4.6211	15.0213	0.6919	418.2820	h24__clf_logen01__reg_xgbr01
2	24	xgb_classifier	xgb_regressor	83.3221	262.0266	3.3876	11.3270	0.8888	418.2820	h24__clf_xgbc01__reg_xgbr01
3	24	xgb_classifier	elasticnet_regression	86.2641	271.3599	0.4456	1.9937	0.8888	446.6630	h24__clf_xgbc01__reg_enet01
4	24	logistic_enet	elasticnet_regression	87.1728	270.6840	-0.4631	2.6696	0.6919	446.6630	h24__clf_logen01__reg_enet01
1	48	xgb_classifier	xgb_regressor	64.3431	187.7245	2.3737	9.0062	0.8740	318.7201	h48__clf_xgbc01__reg_xgbr01
2	48	logistic_enet	xgb_regressor	64.8244	188.5769	1.8923	8.1537	0.6870	318.7201	h48__clf_logen01__reg_xgbr01

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
3	48	xgb_classifier	elasticnet_regression	66.0705	194.9579	0.6463	1.7728	0.8740	334.9688	h48__clf_xgbc01__reg_enet01
4	48	logistic_enet	elasticnet_regression	66.8353	194.9183	-0.1186	1.8124	0.6870	334.9688	h48__clf_logen01__reg_enet01

Winning pairs by horizon

horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage1_f1	mean_stage2_positive_rmse	pair_tag
24	logistic_enet	xgb_regressor	82.0886	258.3323	4.6211	15.0213	0.6919	0.5747	418.2820	h24__clf_logen01__reg_xgbr01
48	xgb_classifier	xgb_regressor	64.3431	187.7245	2.3737	9.0062	0.8740	0.6814	318.7201	h48__clf_xgbc01__reg_xgbr01

Horizon 24h selected Logistic Enet + Xgb Regressor. Average county RMSE expected was 82.0886 versus a zero baseline of 86.7097. Improved by 4.6211 (5.33%).

Horizon 48h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 64.3431 versus a zero baseline of 66.7168. Improved by 2.3737 (3.56%).

Stage-specific summaries

Stage 1 classifiers

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
24	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8888	0.8594	0.1736	0.4183
24	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.6919	0.5777	0.2336	0.5747

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
48	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8740	0.8237	0.1374	0.6814
48	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.6870	0.5544	0.2307	0.5414

Stage 2 regressors

horizon	regressor_name	regressor_params	mean_stage2_positive_rmse	mean_stage2_positive_mae	mean_stage2_positive_rows_train
24	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	418.2820	153.8438	140,860
24	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	446.6630	167.6790	140,860
48	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	318.7201	99.2482	252,291
48	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	334.9688	108.1905	252,291

Submission and refit artifacts

Submission model summary

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
24	Best validation candidate	h24__clf_logen01__reg_xgbr01	1	logistic_enet	xgb_regressor	82.0886	258.3323

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
48	Best validation candidate	h48__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	64.3431	187.7245

Final refit summary

horizon	bundle_source	pair_tag	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected
24	Final refit	h24__clf_logen01__reg_xgbr01	logistic_enet	xgb_regressor	82.0886	258.3323
48	Final refit	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	64.3431	187.7245

Prediction exports

file	rows	prediction_column	min	max	mean	std
two_stage_pred_24h.csv	1,992	pred	1.2532	466.9498	10.2566	35.7400
two_stage_pred_48h.csv	3,984	pred	0.5003	147.1953	5.7420	15.3830
final_refit_pred_24h.csv	1,992	pred	0.8013	583.3138	12.4617	45.2865
final_refit_pred_48h.csv	3,984	pred	0.0085	317.0558	6.9210	26.9863

The predictions folder also contains 8 pair_*_val_predictions.csv file(s) with row-level validation outputs for candidate analysis.

Demo-only evaluation files

These metrics were computed against any demo labels bundled with the repository. They are useful for quick checks, but they are not the official hidden-test challenge results.

file	overall_rmse	overall_mae	avg_county_rmse	n_rows	n_locations
demo_eval_24h.json	37.0404	9.7606	10.6177	1,992	83
demo_eval_48h.json	16.2599	5.2524	5.5135	3,984	83
demo_eval_final_refit_24h.json	46.8303	11.9658	13.4922	1,992	83
demo_eval_final_refit_48h.json	27.7450	6.6037	7.2000	3,984	83

Reusable model artifacts

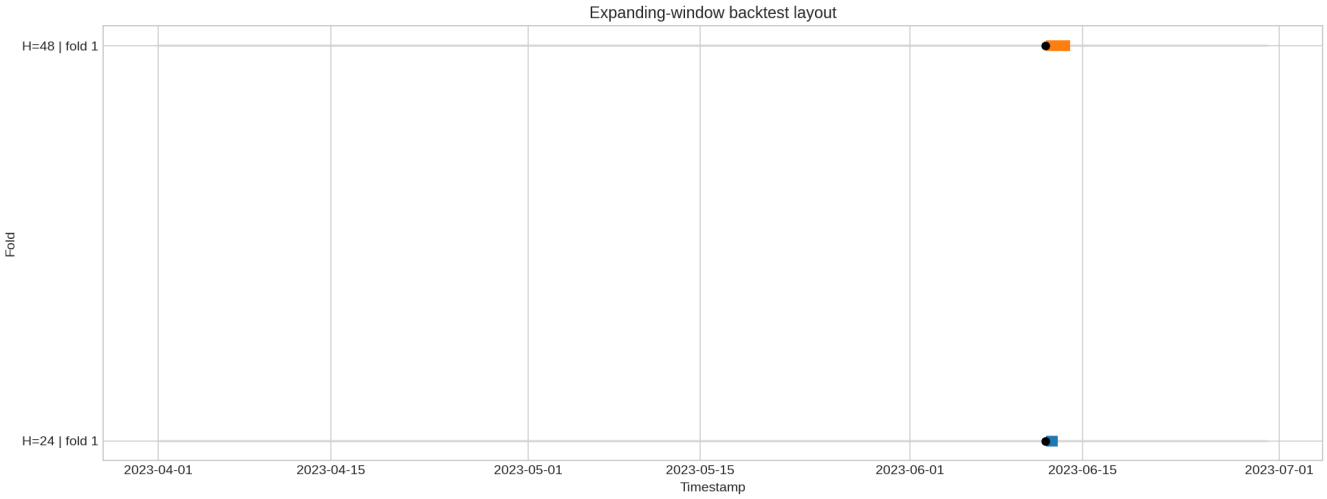
horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
24	1	h24__clf_logen01__reg_xgbr01	logistic_enet	xgb_regressor	pair_h24__clf_logen01__reg_xgbr01_f01_two_stage.joblib	pair_h24__clf_logen01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h24__clf_logen01__reg_xgbr01_f01_stage2_regressor.joblib

horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
48	1	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h48__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

Visual appendix

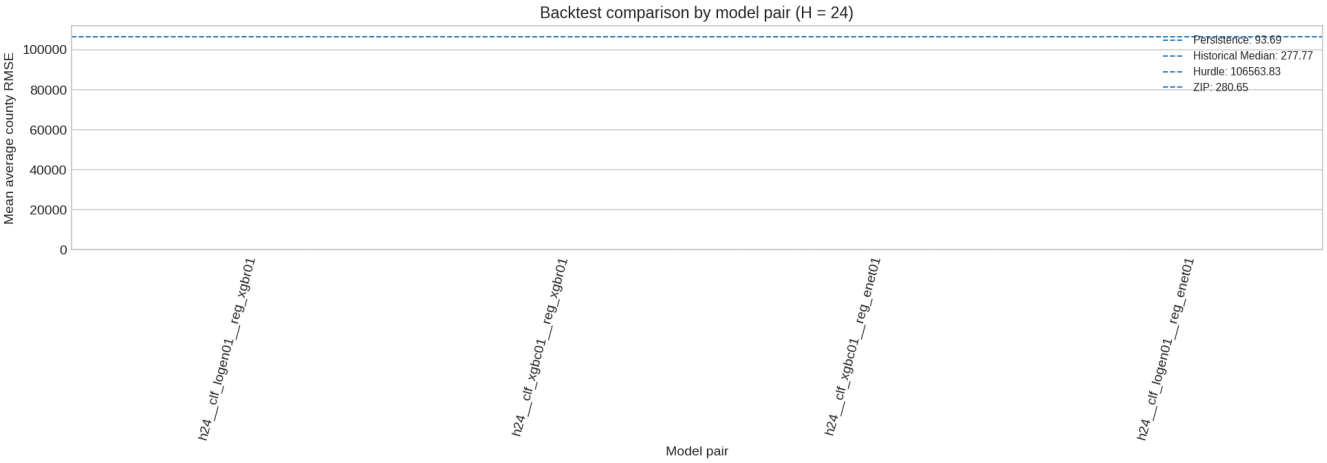
The following figures were read directly from the run's plots folder and inserted without GenAI interpretation.

Backtest Layout



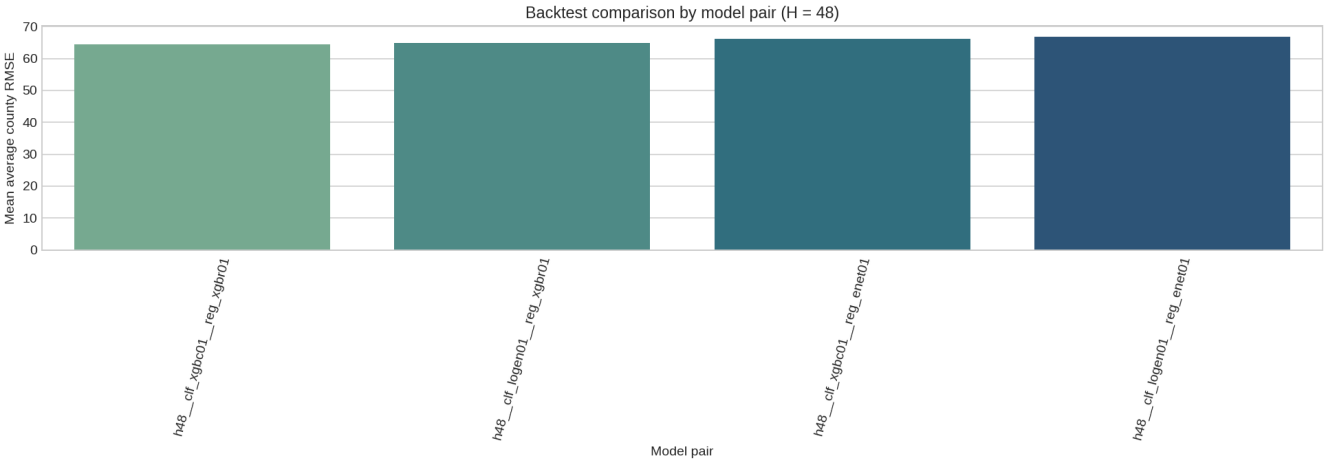
backtest_layout.png

Pair Comparison 24H



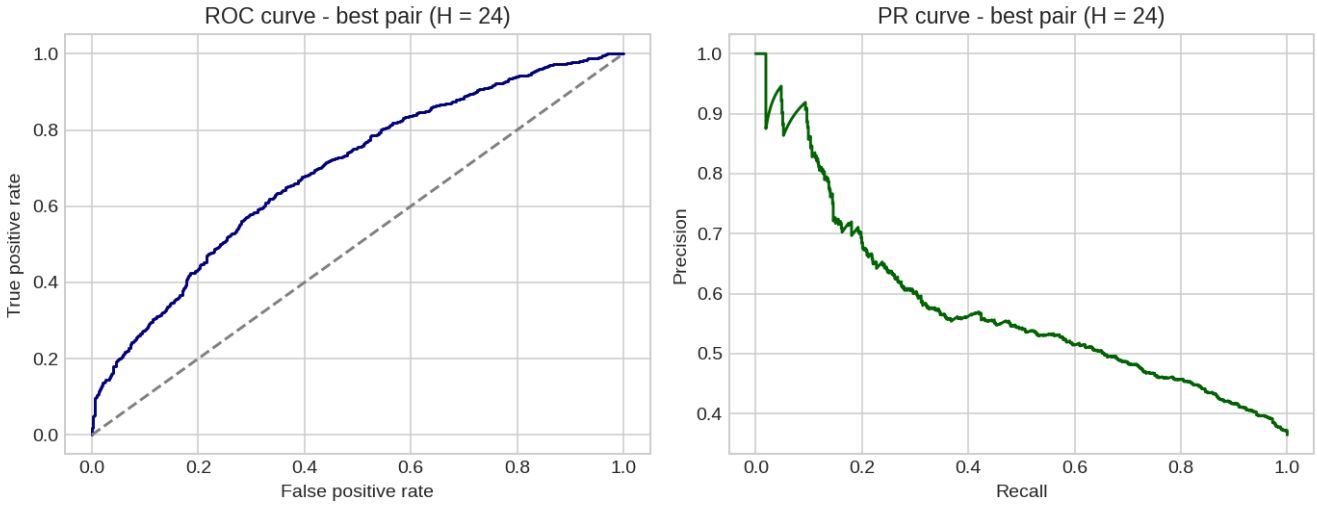
pair_comparison_h24.png

Pair Comparison 48H



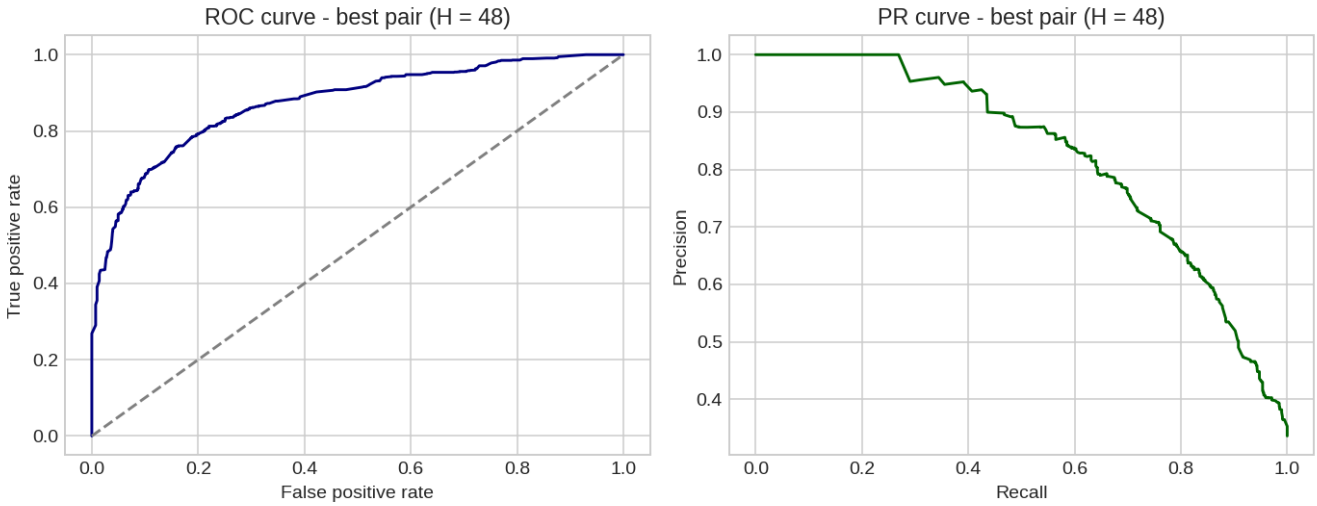
pair_comparison_h48.png

Best Pair Curves 24H



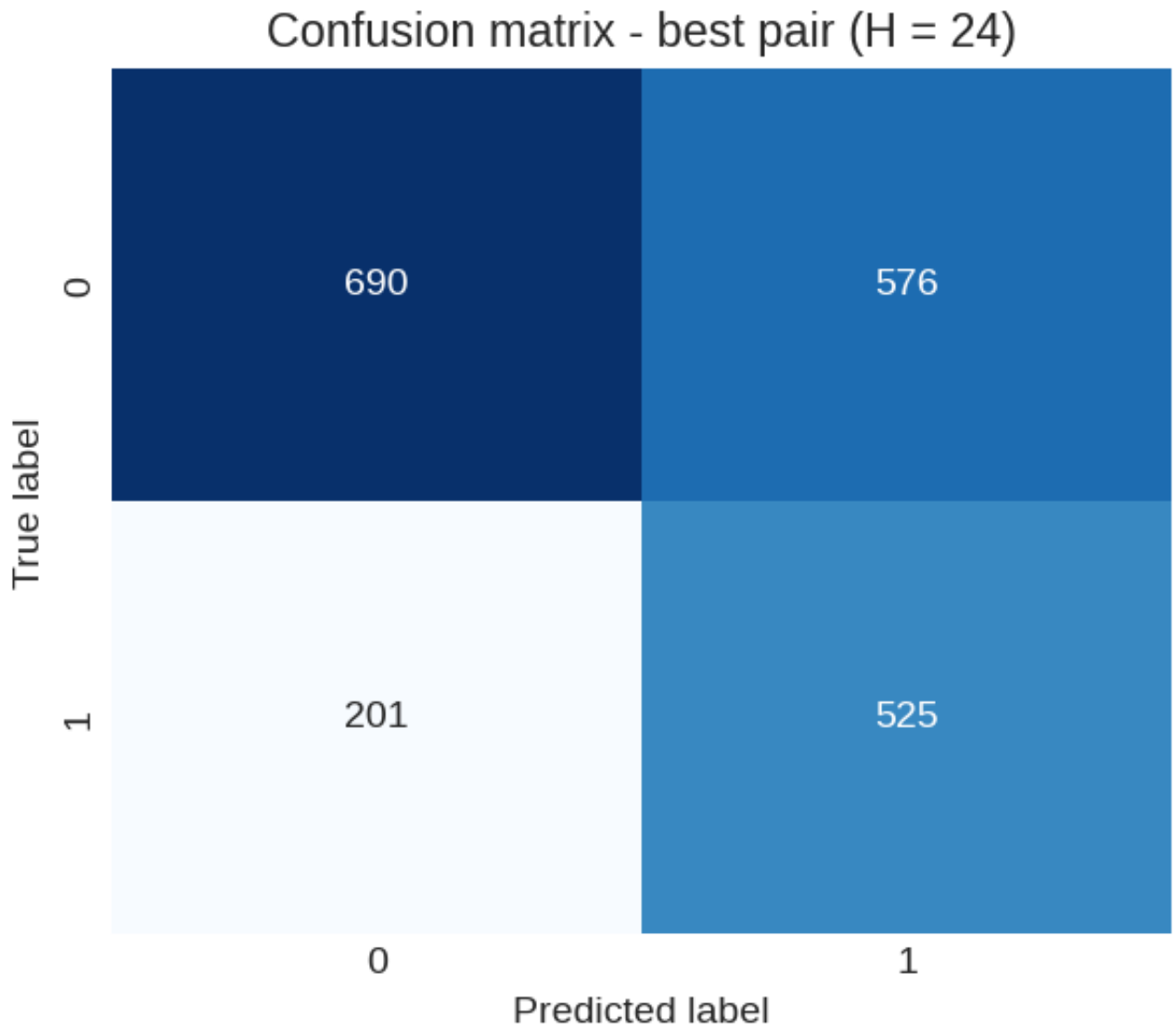
best_pair_curves_h24.png

Best Pair Curves 48H



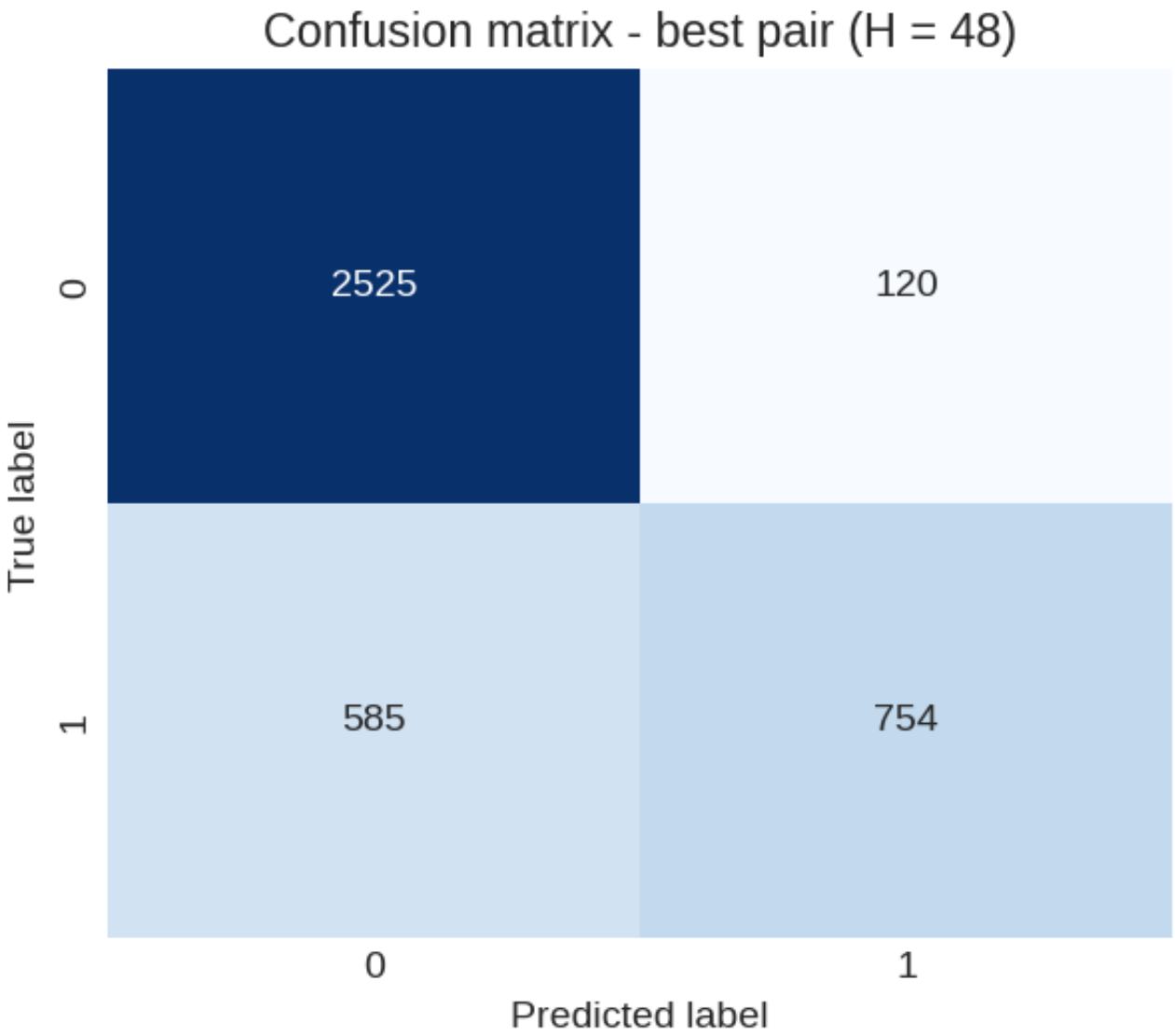
best_pair_curves_h48.png

Best Pair Confusion 24H



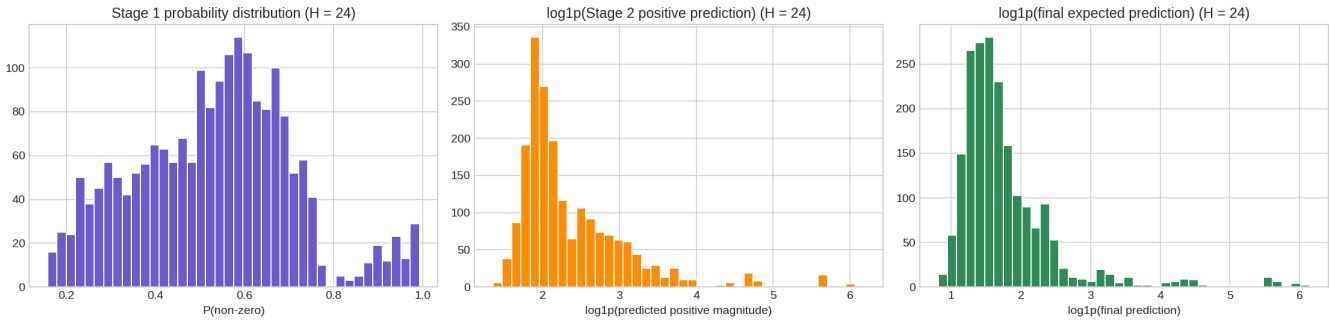
best_pair_confusion_h24.png

Best Pair Confusion 48H



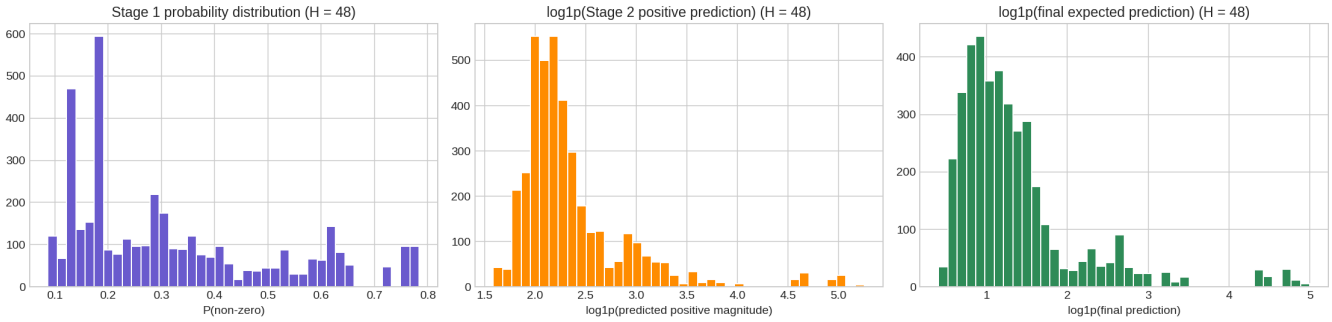
best_pair_confusion_h48.png

Prediction Distributions 24H



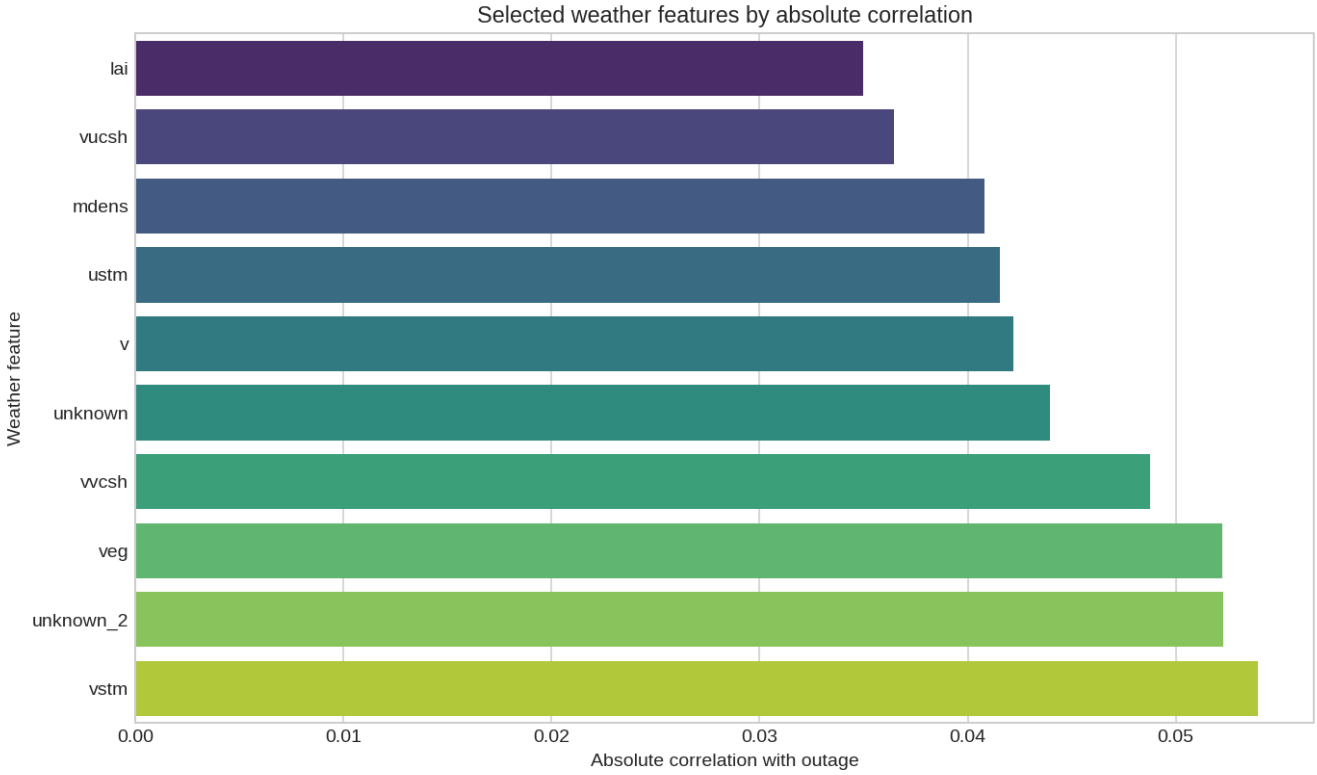
prediction_distributions_h24.png

Prediction Distributions 48H



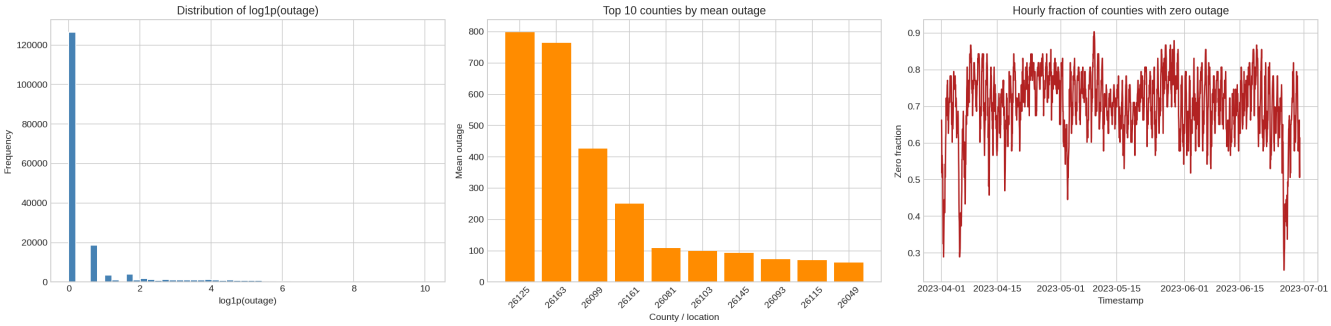
prediction_distributions_h48.png

Selected Weather Features



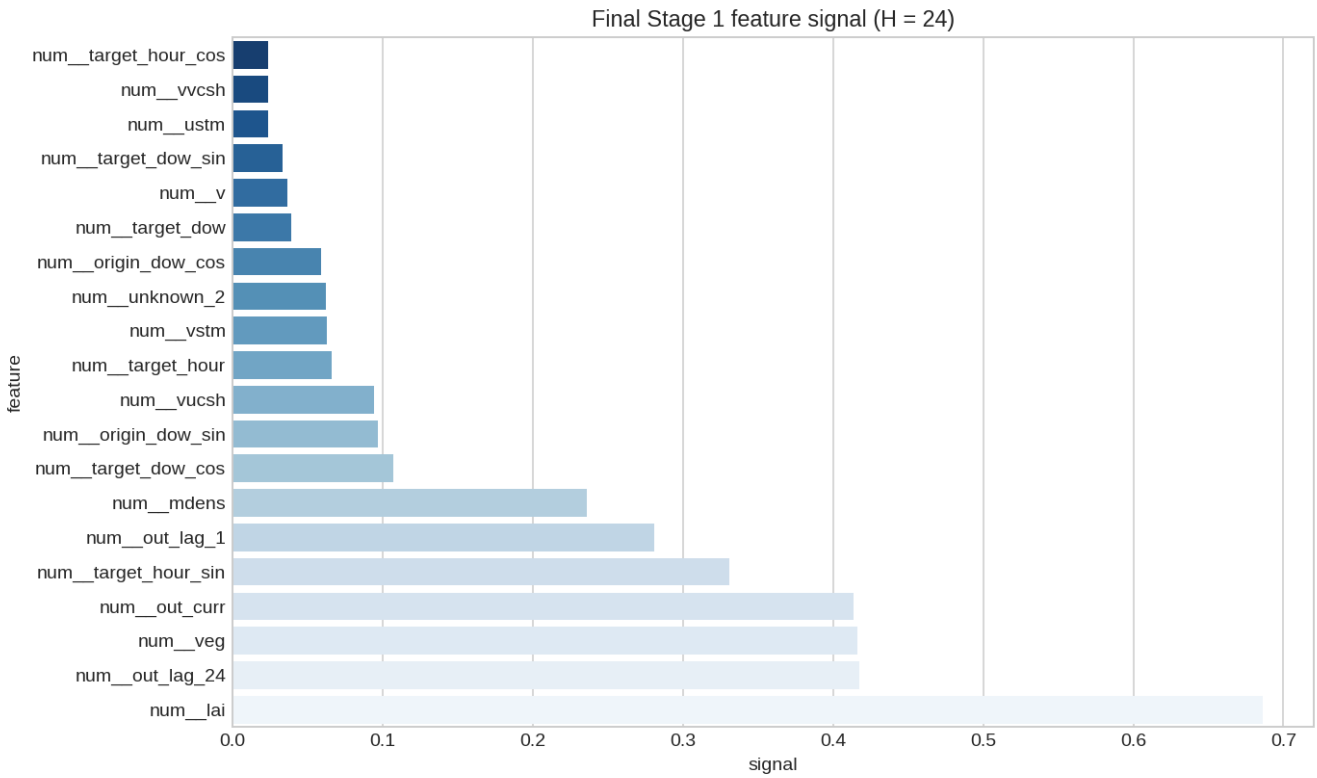
selected_weather_features.png

Eda Overview



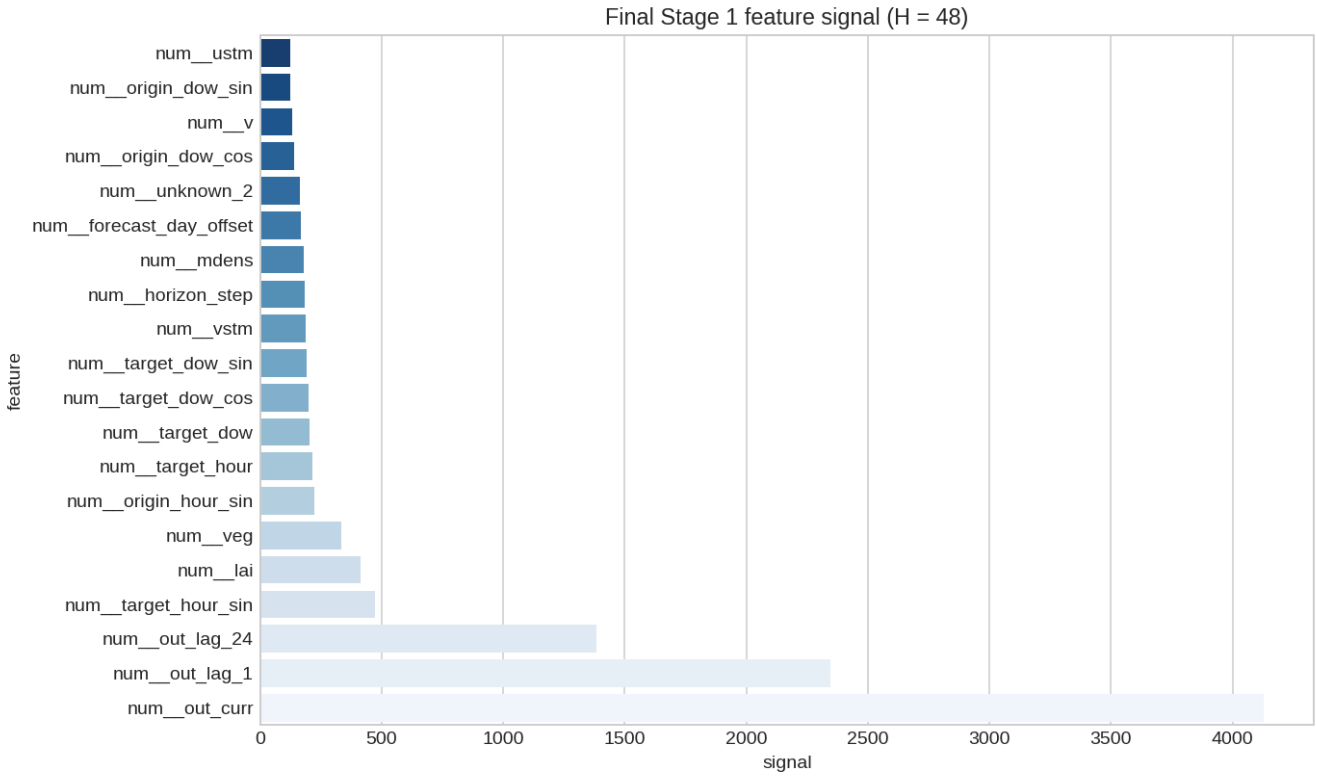
eda_overview.png

Final Stage1 Feature Signal 24H



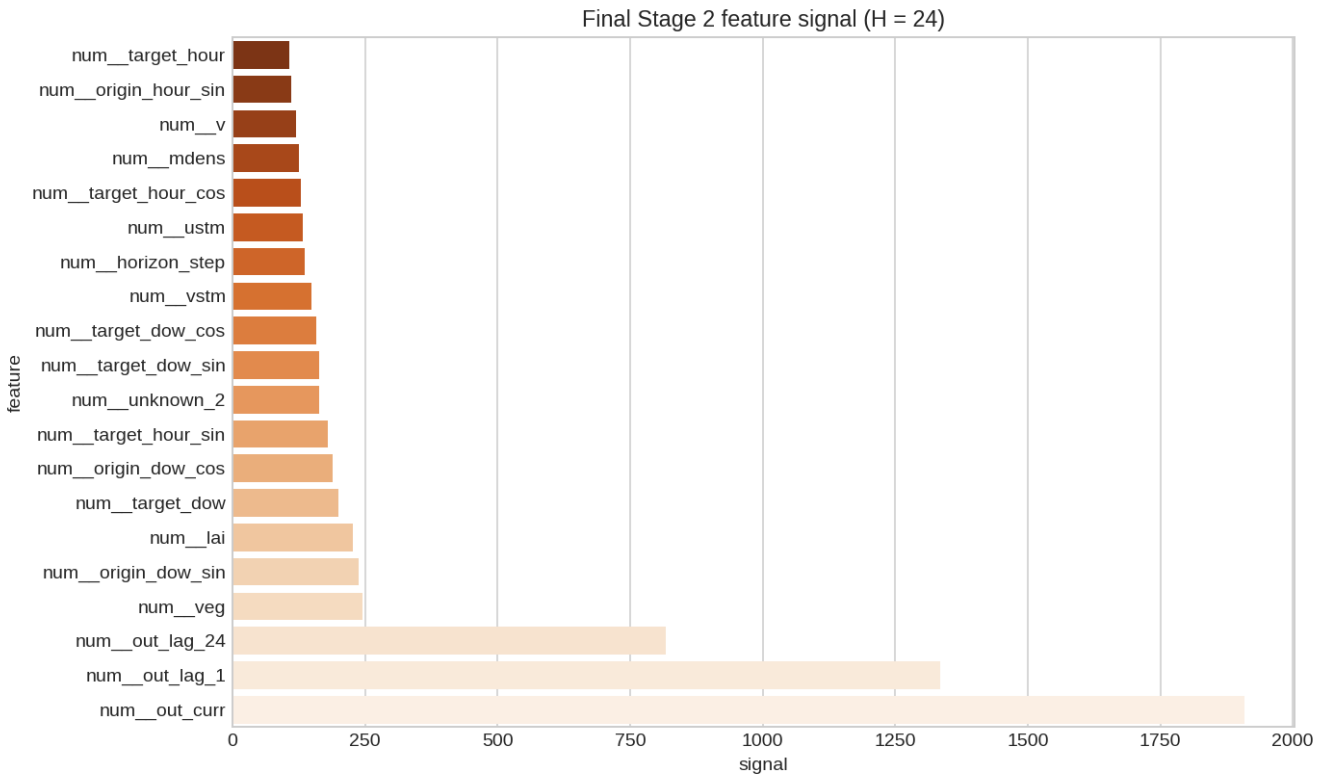
final_stage1_feature_signal_h24.png

Final Stage1 Feature Signal 48H



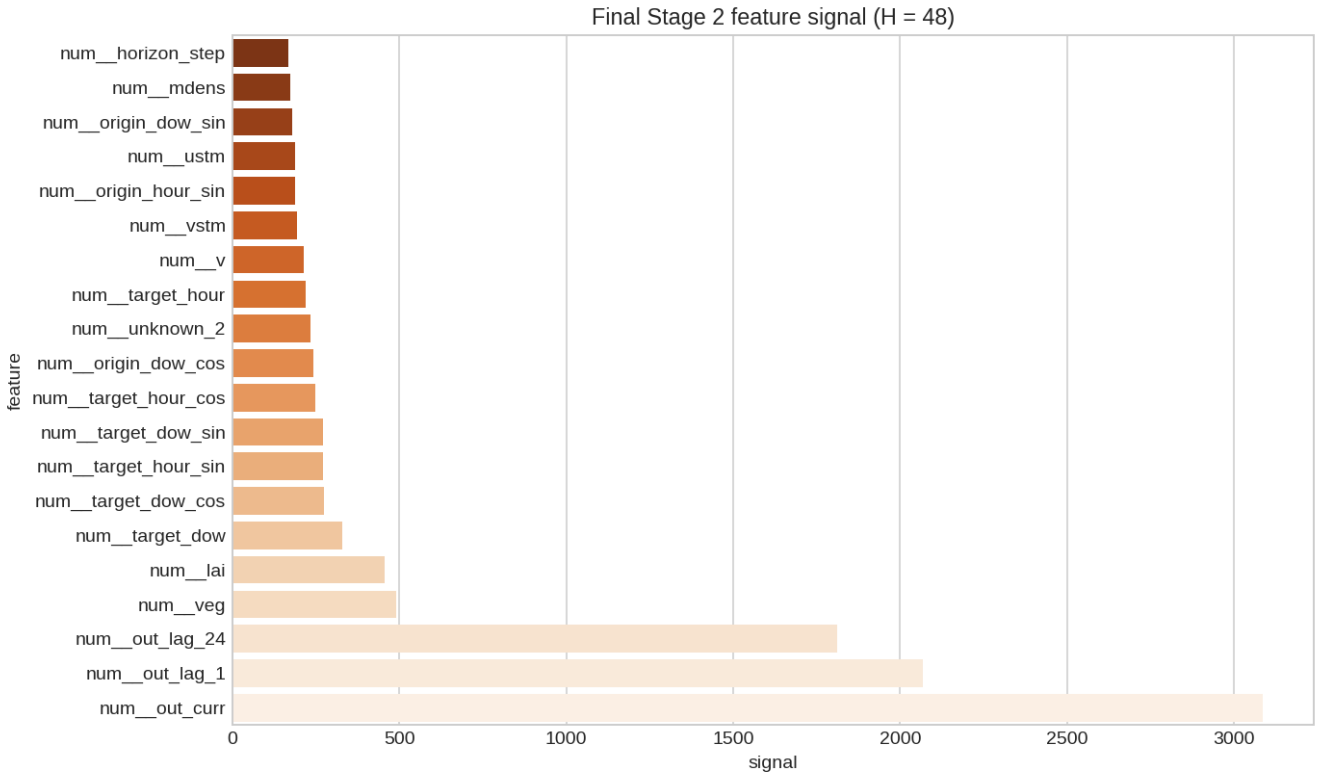
final_stage1_feature_signal_h48.png

Final Stage2 Feature Signal 24H



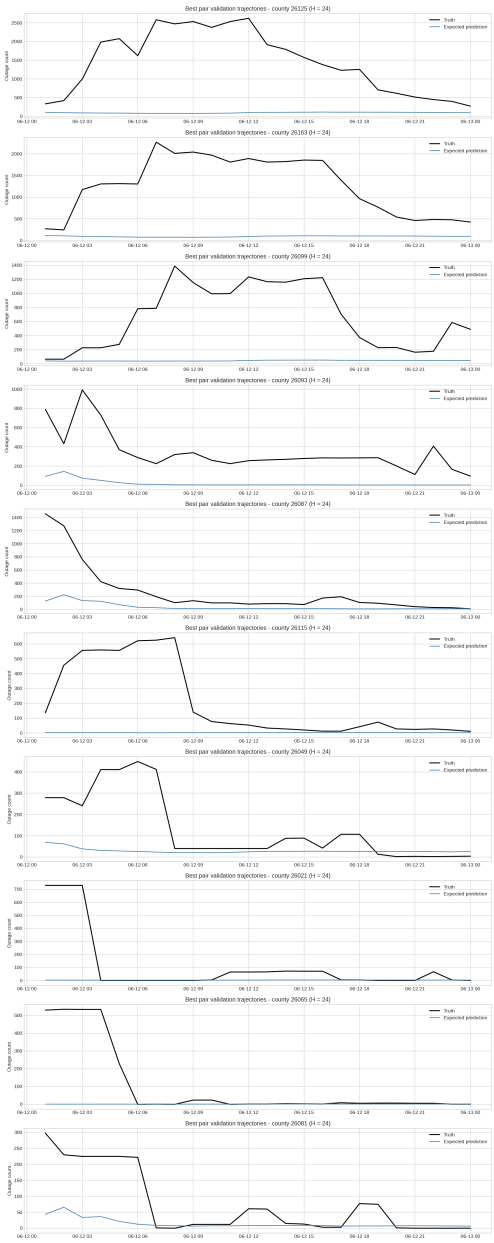
final_stage2_feature_signal_h24.png

Final Stage2 Feature Signal 48H



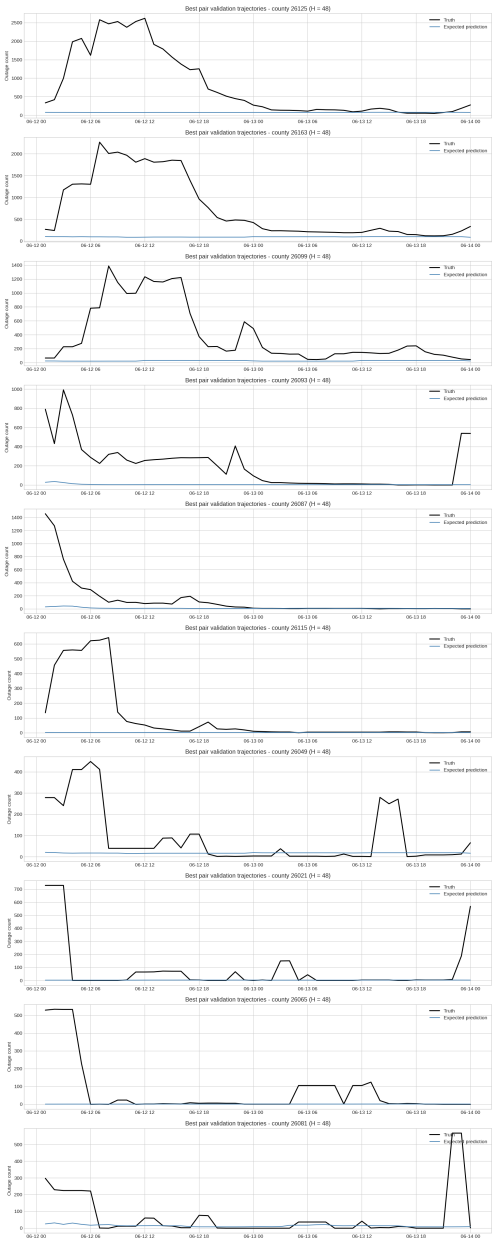
final_stage2_feature_signal_h48.png

Best Pair Validation Trajectories 24H



best_pair_validation_trajectories_h24.png

Best Pair Validation Trajectories 48H



best_pair_validation_trajectories_h48.png

Folder guide

- tables/: aggregated summaries, rankings, split definitions, and run-level CSV records.
- predictions/: exported submission-style predictions and detailed candidate validation predictions.
- models/: reusable .joblib artifacts plus metadata JSON files for submission and refit bundles.
- metrics/: fold-level metrics, training histories, and optional demo evaluation JSON files.
- plots/: visual diagnostics such as split layouts, model comparisons, confusion matrices, and feature-signal plots.
- checkpoints/: runtime resume artifacts used during notebook execution. This folder may be intentionally absent from GitHub uploads.